

- Find the distance between each pair of points.

1) $(0, -8), (-6, 0)$

2) $(-7, -1), (-2, -4)$

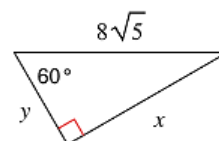
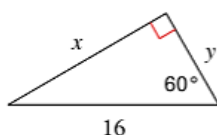
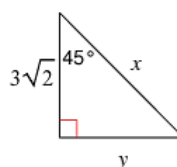
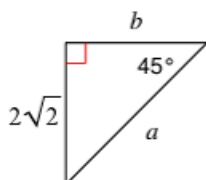
3. Find the midpoint of the segment drawn between the two points. Write your answer as a fraction or decimal rounded to the hundredths place.

$(4, -11)$ and $(12, -5)$

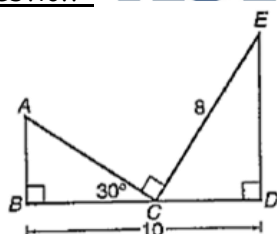
4. Find the midpoint of the segment drawn between the two points. Write your answer as a fraction or decimal rounded to the hundredths place.

$(0, -5)$ and $(9, 7)$

- Find the missing side lengths. Leave your answers as radicals in simplest form.



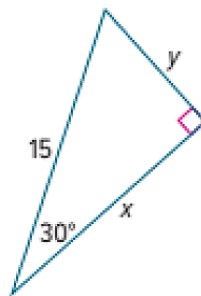
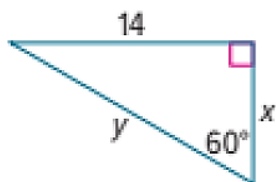
- Bonus Question:



In the figure above, what is the length of line segment AB ?

- (A) 3
(B) $2\sqrt{3}$
(C) 4
(D) $3\sqrt{3}$
(E) 6

- Find the values of x & y :



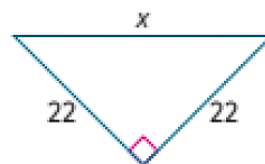
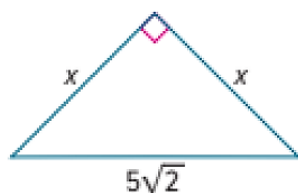
- Find the distance between each pair of points.

$X(-2, 5, -1)$ and $Y(9, 0, 4)$

- Determine the coordinates of the midpoint M :

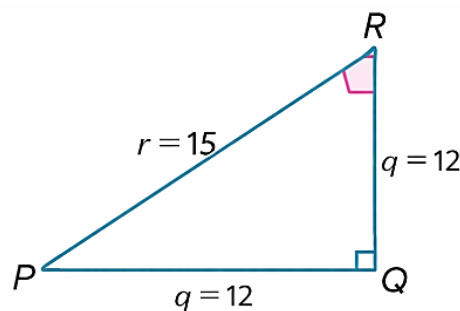
$K(-2, -4, -4)$ and $L(4, 2, 0)$

- Find the value of x :

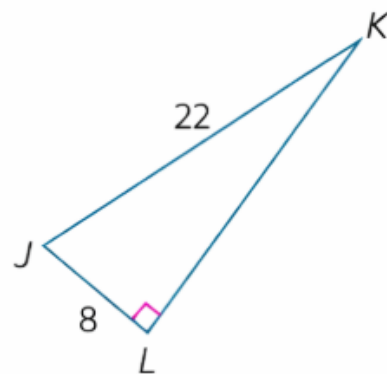
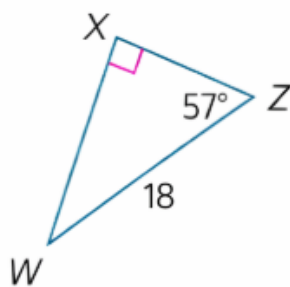
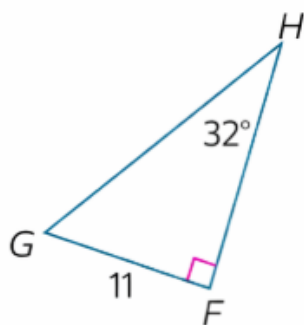


- Find $\sin P$, $\cos P$, $\tan P$, $\sin Q$, $\cos Q$, and $\tan Q$. Express each ratio as a fraction and as a decimal rounded to the nearest hundredth.

1. $p = 9$, $q = 12$, $r = 15$

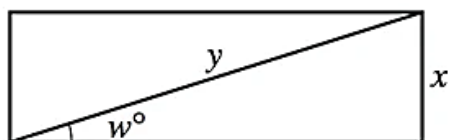


Solve each **right triangle**. Round side measures to the nearest tenth and angle measures to the nearest degree.



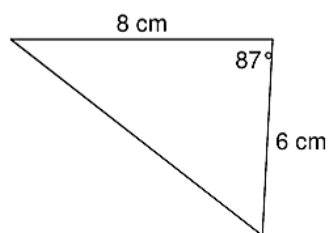
- Bonus Question: **ACT**

Which of the following trigonometric equations is valid for the side measurement x inches, diagonal measurement y inches, and angle measurement w° in the rectangle shown below?

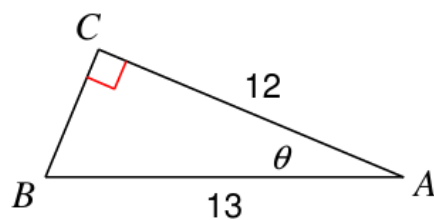


- A. $\cos w^\circ = \frac{x}{y}$
- B. $\cot w^\circ = \frac{x}{y}$
- C. $\sec w^\circ = \frac{x}{y}$
- D. $\sin w^\circ = \frac{x}{y}$
- E. $\tan w^\circ = \frac{x}{y}$

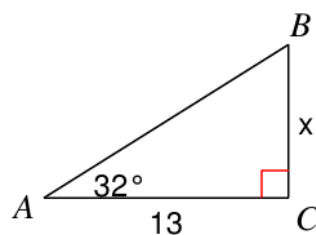
- Find the area of the triangle:



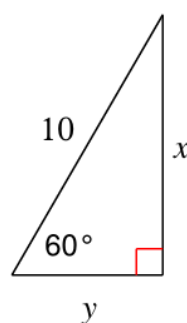
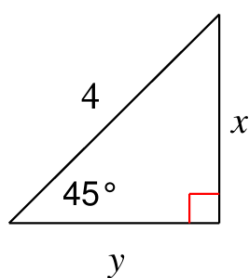
- Find the measure of the angle indicated:



- Find the measure of the side indicated:



- Find the missing side lengths.



- Find the distance:

$(4, 3), (-3, 6)$

- Find the midpoint:

$$(2, -1), (-6, 0)$$

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- From the top of the Niagara Escarpment, Juan sees a car below at an angle of depression (or descent) of 40° . Juan is approximately 100 m above the car. How far is the car from the base of the escarpment, to the nearest metre?



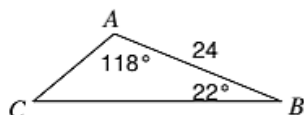
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- Bonus Question: **ACT**[®]

If tangent of an angle is $\frac{1}{2}$, what is the cosine of the same angle?

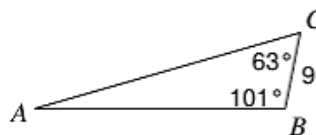
- A. $\frac{5}{2}\sqrt{2}$
- B. $\frac{1}{2}$
- C. $\frac{1}{5}$
- D. $\frac{2\sqrt{5}}{5}$
- E. $\frac{\sqrt{2}}{3}$

- Find each measurement indicated:

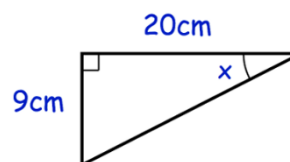
1) Find AC



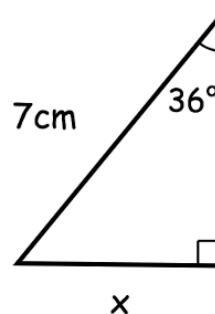
2) Find AB



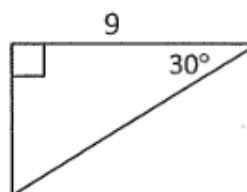
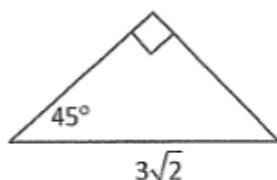
- Find the measure of angle x:



- Find the measure of the value of x:



- Find the missing side lengths for these special right triangles.



- Calculate the distance between the points $(-5, 7)$ and $(-3, -2)$.

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- Find the midpoint for each line segment using the formula (no graphing needed). Show the formula and all work.

$K (1, -1)$ and $L (8, -7)$

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- Find the Area of the triangle:



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- The angle of elevation from a boat to the top of a 90 m hotel is 10° . How far is the boat from the base of the hotel?

-
- Bonus Question:

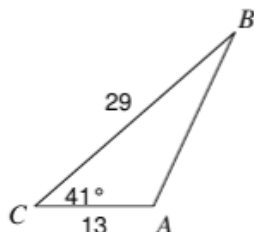
SAT

In a right triangle, one angle measures x° , where

$\sin x^\circ = \frac{4}{5}$. What is $\cos(90^\circ - x^\circ)$?

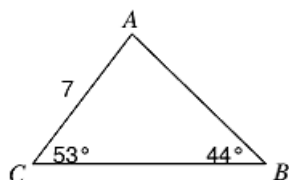
- Find the measurement indicated using Laws of cosines:

1) Find AB

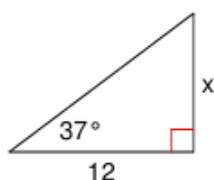


- Find the measurement indicated using Laws of sines:

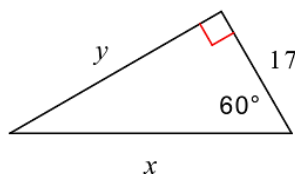
2) Find AB



- Find the measure of the value of x using trigonometry:



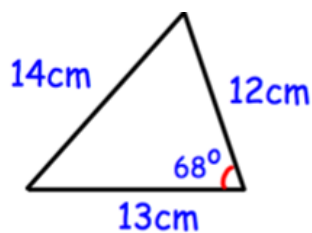
- Find the missing side lengths for the special right triangle.



- Find the midpoint for each line segment using the formula (no graphing needed). Show the formula and all work.

$K(-5, 2)$ and $L(5, -4)$

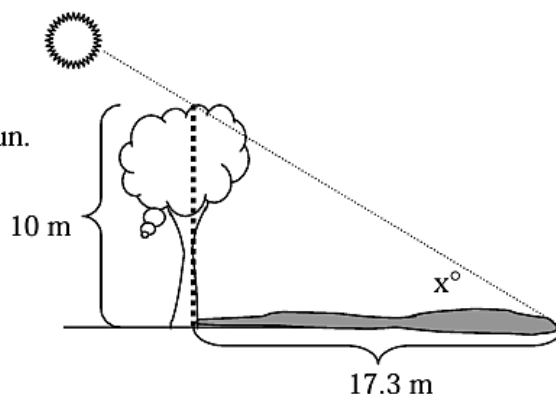
- Find the Area of the triangle:



- State the number of solutions that can be formed using the given measurements, then solve the triangle (Law of Sines Ambiguous Case)

1) $m\angle A = 110^\circ$, $c = 19$ cm, $a = 32$ cm

A tree 10 meters high casts a 17.3 meter shadow. Find the angle of elevation of the sun.



- Bonus Question: **ACT**

In the right triangle below, the measure of $\angle C$ is 90° , $AB = 5$ units, and $CB = 2$ units. What is $\tan B$?

- F. $\frac{\sqrt{21}}{2}$
 G. $\frac{3}{2}$
 H. $\frac{\sqrt{21}}{5}$
 J. $\frac{3}{5}$
 K. $\frac{2}{5}$

